

DATA CARD

Latent Probing for Toxicity Detection

Team Name: Team OKU | Team Leader: Thaabit Idris
Team Members: Ogwu Naomi Chukwufumnanya, Riyas Yusuf
Cohort: AI Saturdays Lagos - Cohort 10
Competition: Latent Probe Challenge - Toxicity Detection (Codabench)
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1. Input vs. Output

Model Input:

Users input text - specifically prompts or requests directed at an AI assistant. This text is tokenised and passed through the Gemma 2-2B language model (google/gemma-2-2b). The model then extracts a 2,304-dimensional embedding vector from layer 14, representing the mean-pooled hidden state across all tokens, truncated at 64 tokens. This vector encodes the model's internal understanding of the input before any response is generated.

Model Output:

A binary prediction per example: 1 (toxic or harmful) or 0 (safe). The probe classifies whether the underlying text represents a harmful or toxic request to an AI assistant.

2. Classification Task

This is a binary classification task. Labels are defined as follows:

Label	Value	Definition
Toxic / Harmful	1	Requests for harmful instructions, jailbreak attempts, hate speech, phishing, PII extraction, self-harm ideation
Safe / Benign	0	Normal conversational prompts, and queries that superficially resemble harmful requests but carry no harmful intent (hard negatives)

Labels from source datasets were mapped to this binary scheme: hate and abusive classes mapped to 1, neutral and benign classes mapped to 0, consistently across all source datasets.

3. Labelling Strategy (updated)

All labels are inherited from existing human-annotated datasets, with two corrections made during development after auditing each source's actual label semantics rather than trusting documented column names at face value:

Dataset	Labelling Method
lmsys/toxic-chat	Real user messages from a deployed chatbot. Note: the dataset's human_annotation column is a review-status flag, not a toxicity label; the correct label is the toxicity column, verified by cross-tabulating both fields before use.
onepaneai/harmful-prompts	Expert-curated harmful instruction prompts covering jailbreaks, prompt injection, and general toxicity; all examples toxic by construction.

PKU-Alignment/BeaverTails	Human preference annotations; binary is_safe field. Prompts repeat across multiple response/category rows (mean 3.5 duplicates per unique prompt); deduplicated to 7,766 unique prompts, with label conflicts across duplicates (32% of unique prompts) resolved by majority vote.
OpenAssistant/oasst1	Community-annotated conversations. Used exclusively as a safe-class supplement: English, user-authored (role=prompter) messages with an automated Detoxify toxicity score at or below 0.5. Manual review found Detoxify's score reflects crude/explicit language rather than harmful-request intent, so the small subset scoring above 0.5 was excluded rather than treated as this task's toxic class.

WildJailbreak (allenai/wildjailbreak) was evaluated and is not part of the final training set - see Section 4.

4. Data Creation (rewritten)

Five candidate sources were evaluated empirically rather than assumed suitable. Two were tested and excluded based on measured impact on held-out accuracy, not on assumption:

- allenai/wildjailbreak was excluded. Its synthetic, template-generated prompts made up 94% of the combined pool at full size, and every training combination that included it scored lower on the competition's hidden test set than the same combination without it. We interpret this as a distribution mismatch between WildJailbreak's synthetic construction and the style of toxicity the hidden test set actually contains.

- google/civil_comments (public internet comment toxicity, added as a candidate during development) was tested and excluded for the same reason: it measures incivility (rudeness, profanity) rather than harmful requests to an AI assistant, and reduced accuracy in every combination tested.

Dataset	Purpose	Rows Used
lmsys/toxic-chat	Toxic + safe class - real conversational messages	5,082
onepaneai/harmful-prompts	Toxic class - jailbreak, injection, general harmful prompts	200
PKU-Alignment/BeaverTails	Toxic + safe class, wide harm-category coverage	7,766 (deduped)
OpenAssistant/oasst1	Safe class - realistic, non-adversarial requests	3,000 (sampled)

Total: 16,048 examples (4,862 toxic / 11,186 safe). Class imbalance is addressed at training time via class-weighted logistic regression rather than resampling.

BeaverTails role change:

Originally planned as a held-out out-of-distribution validation set only. Once WildJailbreak was excluded, retaining BeaverTails as unused validation data left the training set too small and narrow; it was moved into training (deduplicated first). This means our internal held-out accuracy figure is no longer independently validated against a fully unseen source - the competition's own hidden test set is now the only genuinely independent validation available, and our reported score comes from that, not from an internal out-of-distribution set.

African-language layer: planned, not completed.

The original plan called for translating a sample of examples into Hausa and Igbo via NLLB-200 and constructing Nigerian Pidgin examples manually, with Yoruba held out as a documented evaluation gap. This work did not happen. Development time was consumed by a substantial debugging effort: two data-labelling bugs (Section 3) and a layer-extraction bug (a forward hook was registered one transformer layer off from the competition's specified layer, discovered only after comparing self-consistent local validation against the platform's genuinely independent scoring) took priority as blocking issues, and the remaining time before the Sept 9 development-phase deadline went to testing

and validating additional English-language sources instead. **The final probe was trained and evaluated on English-language data only.** This is a real limitation, not a hidden one - see Section 6.

5. Web Collection Ethics

No web scraping was performed. All datasets were obtained from Hugging Face or GitHub under their published open-source licences. No personally identifiable information was collected.

WildJailbreak is a gated dataset. Access was formally requested and granted before use, in compliance with the dataset authors' terms of service, prior to its exclusion from the final training set following empirical evaluation (Section 4).

Translated examples were not produced in the final submission (Section 4). No AI tools were used to generate labels - all labels are inherited from the original human-annotated source datasets.

Copyright over source dataset content remains with the original authors. Our use is for non-commercial academic research within the AI Saturdays Lagos Cohort 10 programme.

6. Bias and Representation (updated)

Who might be left out:

Language coverage - full gap, not partial:

The final training set is 100% English. The African-language translation layer described in the original Data Card was not built (Section 4). This is a more significant limitation than originally documented, not a smaller one: rather than an English-majority probe with a translated minority and a single documented Yoruba gap, this is an English-only probe with no verified performance data in Nigerian Pidgin, Hausa, Igbo, or Yoruba. We do not know how the probe behaves on non-English input, and did not have time to test.

Cultural context (unchanged):

The toxic class is defined largely by Western AI safety research norms. Concepts of harm and acceptable speech vary across Nigerian communities and are not captured by our label taxonomy.

Class construction (revised):

The original plan relied on WildJailbreak's synthetic hard-negative benign split to teach the probe to distinguish genuine harmful intent from superficially sensitive language. With WildJailbreak excluded, this specific mitigation is no longer part of the pipeline; BeaverTails and oasst1's real (non-synthetic) prompts are the primary safe-class source instead, which may reduce hard-negative coverage for edge cases WildJailbreak specifically targeted.

How we are improving fairness and representation, revised to reflect what actually happened:

- Language coverage is documented as a full gap rather than partially addressed. We are not claiming Hausa, Igbo, or Pidgin support in this submission.
- Every dataset inclusion/exclusion decision was tested empirically against held-out accuracy (Section 4) rather than assumed, including catching two label-definition bugs that would otherwise have silently trained the probe on incorrect signal.
- We are publishing this limitation explicitly, in line with the transparency commitment in Section 7, rather than adjusting the description to imply broader coverage than what was delivered.

7. Values and Community Alignment (updated)

Our dataset design is guided by five values drawn directly from our Challenge 1 problem statement: equity, accessibility, transparency, proportionality and linguistic fairness.

Equity and accessibility:

Every dataset used is openly licensed and freely available, reflecting our commitment that safety research should not require expensive proprietary data. The probe reuses computation the model already performs, making safety accessible to African developers who cannot afford commercial moderation APIs or second guardrail models.

Transparency:

We report dataset composition, known bias sources, and the full reasoning behind every inclusion and exclusion decision, including decisions that diverged from our original plan. Rather than quietly updating this document to imply the original scope was delivered, we name each change and the reasoning behind it directly.

Proportionality:

Our safe class draws on real (non-synthetic) prompts from BeaverTails and oasst1 to help the probe distinguish genuine harmful intent from superficially sensitive but legitimate speech, addressing the risk of a safety tool that suppresses ordinary conversation.

Linguistic fairness - revised:

Our original commitment was to build a probe that serves Nigerian Pidgin, Hausa, and Igbo speakers directly, not just English speakers. We did not deliver on this within the development-phase timeline. The honest version of this section is: linguistic fairness remains an unmet goal of this submission, not a partially-met one. We consider this the most significant gap between our Challenge 1 problem statement and what we actually built, and name it directly here rather than in a buried limitations footnote, because the communities named in our own problem statement are exactly the ones this gap affects.

Community benefit and accountability:

- Failure and scope changes are documented honestly, including the unmet linguistic fairness goal, rather than hidden or minimised.
- A limitations statement is published that is clear enough for a developer to decide not to deploy our probe where it does not fit their context - in particular, for any non-English deployment.
- Methodology, label taxonomy, and the full record of what was tested and why it was included or excluded are made available so other teams can interrogate, challenge, and improve on our work rather than accepting it as a sealed system.